

Telecom Customer Churn Analysis Using Excel

1. Project Overview

Objective

The objective of this project was to analyze customer churn data from a telecom company to identify factors contributing to customer attrition and provide actionable recommendations to improve customer retention.

2. Dataset Overview

Dataset Information

- Total Customers: 7,043
- Active Customers: 5,174
- Churned Customers: 1,869
- Overall Churn Rate: 26.5%

Key Variables Analyzed

- Contract Type
- Payment Method
- Internet Service
- Tech Support
- Senior Citizen Status
- Online Security
- Monthly Charges
- Customer Lifetime Value (CLTV)

3. Tools Used

Microsoft Excel

- Data Cleaning
- Pivot Tables
- Pivot Charts
- KPI Calculations
- Dashboard Design
- Data Visualization

4. Analysis Process

Step 1: Data Understanding

Reviewed customer demographics, service subscriptions, billing information, and churn indicators to understand the dataset structure.

Step 2: Exploratory Analysis

Used Pivot Tables to compare churn behavior across different customer segments.

Step 3: KPI Development

Calculated key business metrics:

- Total Customers
- Churned Customers
- Churn Rate
- Average Monthly Charges
- Average CLTV

Step 4: Dashboard Creation

Built an interactive dashboard to visualize churn trends and highlight key risk segments.

5. Key Findings

A. Contract Type is the Strongest Churn Driver

Customers on month-to-month contracts experienced a churn rate of **42.7%**, significantly higher than one-year (**11.3%**) and two-year (**2.8%**) contracts.

B. Electronic Check Customers Show Highest Churn

Customers using electronic check payments churned at **45.3%**, suggesting payment behavior may be associated with retention risk.

C. Fiber Optic Customers Have Elevated Churn

Fiber optic customers exhibited a churn rate of **41.9%**, more than double that of DSL customers (**19.0%**).

D. Lack of Tech Support Increases Churn

Customers without tech support churned at **41.6%**, compared to **15.2%** for customers receiving support services.

E. Senior Citizens Are More Likely to Churn

Senior citizens showed a churn rate of **41.7%**, compared to **23.6%** among non-senior customers.

F. Online Security Improves Retention

Customers without online security churned at **41.8%**, while those subscribed to online security services churned at only **14.6%**.

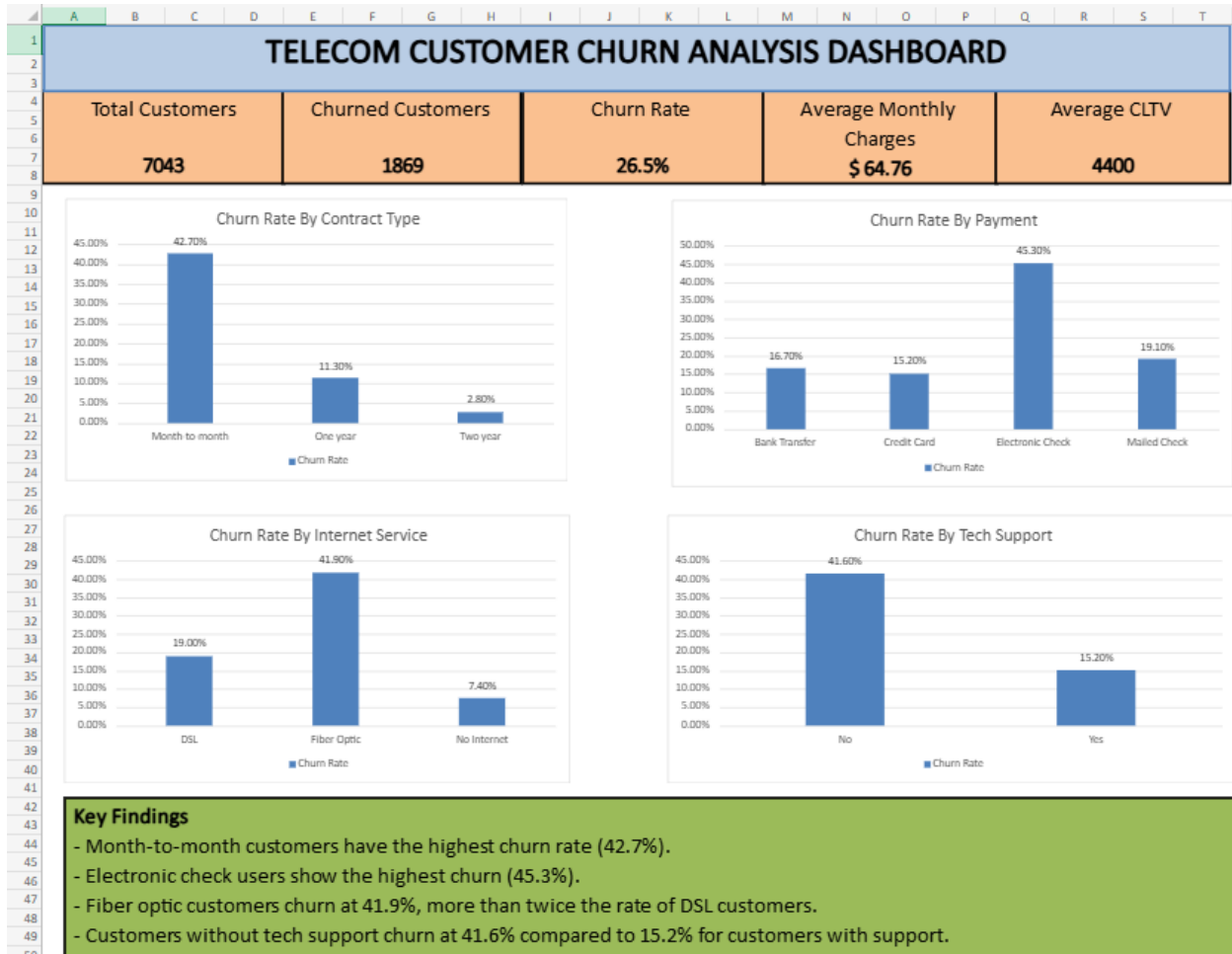
6. Business Recommendations

Customer Retention Strategy

- Encourage customers to transition from month-to-month contracts to longer-term plans.
- Promote automatic payment methods to reduce churn risk.
- Investigate service quality and customer experience issues among fiber optic customers.
- Bundle tech support and online security services into retention-focused packages.
- Develop targeted engagement programs for senior citizen customers.

7. Project Outcome

This analysis identified key customer segments with elevated churn risk and highlighted opportunities for improving retention through contract optimization, service enhancements, and customer support initiatives.



Excel Dashboard